



Equivalent forms of rational numbers

Teacher resource

Question 1

Which fraction has the same value as $2\frac{3}{4}$?

$$\frac{8}{4}$$

☐

$$\frac{9}{4}$$

☐

$$\frac{11}{4}$$

☐

$$\frac{14}{4}$$

☐

Answer: C

Skill: Students recognise and convert mixed numerals into improper fractions.

Background information

When thinking of fractions as quantities, students should understand that the same fractional quantity can be represented using fractions with different denominators. When thinking of fractions as points on a number line, students should understand that a point can be named in different ways, including fractions with different denominators and decimal fractions.

Additional questions

1. Write $1\frac{1}{2}$ as an improper fraction.
2. Write $\frac{7}{5}$ as a mixed number.

Question 2

A farmer has five sheep. Three are black.

What percentage of the farmer's sheep is black?

$$3\%$$

☐

$$30\%$$

☐

$$35\%$$

☐

$$60\%$$

☐

Answer: D

Skill: Students convert fractions to percentages.

Additional questions

1. Write a percentage equivalent to one eighth.
2. Write a fraction equivalent to 45%.

Question 3

Choose the smallest number.

$$\frac{2}{5}$$

O

45%

O

0.42

O

$$\frac{4}{9}$$

O

Answer: A

Skill: Students move between equivalent forms of rational numbers.

Background information

Students need to understand that a fraction can be thought of as a division. For example, if five identical items are shared equally between eight people, each person will receive $\frac{5}{8}$ of an item. The result of dividing five by eight is therefore $\frac{5}{8}$. When five is divided by eight on a calculator, the result can be understood as $\frac{5}{8}$ expressed as a decimal.

Additional questions

1. Write $\frac{1}{10}$ as a decimal.
2. Write 30% as a decimal.
3. Write $1\frac{3}{5}$ as a decimal.

Question 4

Which fraction is equivalent to 0.05%?

$$\frac{5}{1000}$$

O

$$\frac{1}{2000}$$

O

$$\frac{5}{20000}$$

O

$$\frac{5}{100000}$$

O

Answer: B

Skill: Students convert percentages to fractions

Additional questions

1. What is $\frac{5}{1000}$ as a percentage?
2. What is .002 as a percentage?
3. What is $1\frac{1}{100}$ as a percentage?
4. What is 2.1 as a percentage?

Student activity: Equivalent forms of rational numbers

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